

2021 级硕士研究生矩阵分析期末试题

座号 _____ 学院 _____ 学号 _____ 姓名 _____

(试卷共 7 页, 八道大题, 解答题必须有解题过程, 试卷后面空白页撕下做稿纸, 试卷不得拆散)

题号	一	二	三	四	五	六	七	八	总分
得分									
签名									

一、填空题 (每空 3 分, 共 30 分)

1、设 $A(\lambda) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \lambda^2 - 1 & 0 \\ 0 & 0 & \lambda + 2 \end{bmatrix}$, 则 $A(\lambda)$ 的不变因子为 $d_1(\lambda) = 1, d_2(\lambda) = 1$, $d_3(\lambda) = (\lambda+1)(\lambda-1)(\lambda+2)$

$A(\lambda)$ 的初等因子为 $\lambda+1, \lambda-1, \lambda+2$

2、设 $A = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$, 则 $A^{2022} - A^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

3、设 α 为三维列向量, α^T 是 α 的转置, 若 $\alpha\alpha^T = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$, 则 $\alpha^T\alpha = 3$

4、已知矩阵 $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 6 & 2-i \\ 0 & 2+i & 2 \end{bmatrix}$, 则矩阵 A 的谱范数 $\|A\|_2 = 7$, 矩阵 A 的列和范数

$\|A\|_1 = 6 + \sqrt{5}$ 矩阵函数 e^{At} 的行列式值 $|e^{At}| = e^{9t}$, 这里 i 为虚数单位, $i^2 = -1$.

5、已知三阶单位矩阵 I , 则矩阵函数 $e^{\frac{\pi}{2}I} = iI$, 这里 i 为虚数单位, $i^2 = -1$.

6、已知函数矩阵 $A(x) = \begin{bmatrix} \sin x & -e^{2x} \\ e^{2x} & \cos x \end{bmatrix}$, 则 $\frac{d^2 A(x)}{dx^2} = \begin{bmatrix} -\sin x & -4e^{2x} \\ 4e^{2x} & -\cos x \end{bmatrix}$

$\frac{d}{dx} \left(\int_0^x A(t) dt \right) = \begin{bmatrix} 2x \sin x^2 & -2x e^{2x^2} \\ 2x e^{2x^2} & 2x \cos x^2 \end{bmatrix}$

二、(12分) 已知矩阵 $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$.

(1) 求矩阵 A 的最小多项式.

(2) 求矩阵函数 $\cos \frac{\pi}{2} A$ 和 e^{2A} .

A 的 Jordan 标准形为 $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$m_A(\lambda)$ 最小多项式为 $\lambda^2(\lambda-1)^2 \rightarrow 4 \frac{1}{2}$

$f(x)$ 为任意函数

$f(A) = \begin{bmatrix} f(1) & 0 & 0 & 0 \\ f'(1) & f(1) & 0 & 0 \\ 0 & 0 & f(0) & 0 \\ 0 & 0 & f'(0) & f(0) \end{bmatrix} \rightarrow 8 \frac{1}{2}$

$f(x) = \cos \frac{\pi}{2} x$

$f(A) = \cos \frac{\pi}{2} A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ -\frac{\pi}{2} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$f(x) = e^{2x}$

$f(A) = e^{2A} = \begin{bmatrix} e^2 & 0 & 0 & 0 \\ 2e^2 & e^2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 1 \end{bmatrix} \rightarrow 12 \frac{1}{2}$

三、(12分) 已知 $A = \begin{bmatrix} 2 & 0 \\ 0 & i \\ 0 & 0 \end{bmatrix}$, 求矩阵 A 的奇异值分解表达式, 这里 i 为虚数单位, $i^2 = -1$.

$$AA^H = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

A 的奇异值为 $\delta_1 = 2, \delta_2 = 1 \rightarrow 4$ 分

$$\Delta = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\lambda_1 = 4 \rightarrow \alpha_1 = [1, 0, 0]^T$$

$$\lambda_2 = 1 \rightarrow \alpha_2 = [0, 1, 0]^T$$

$$U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\lambda_3 = 0 \rightarrow \alpha_3 = [0, 0, 1]^T$$

$$U^H A A^H U = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow 8 \text{分} \quad U_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$V_1 = A^H U_1 \Delta^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}$$

$$V = [V_1, V_2] = \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix} \rightarrow 10 \text{分}$$

$$A = U D V^H = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \rightarrow 12 \text{分}$$

四、(10分) 已知可对角化矩阵

$$A = \begin{bmatrix} -3 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -3 \end{bmatrix}$$

求矩阵 A 的谱分解.

$$|\lambda I - A| = (\lambda + 1)(\lambda + 3)(\lambda + 4) \quad \lambda_1 = -1, \lambda_2 = -3, \lambda_3 = -4$$

$$\lambda_1 = -1 \rightarrow \alpha_1 = [1, 2, 1]^T, \quad \xi_1 = \frac{1}{\sqrt{6}} [1, 2, 1]^T \rightarrow 2 \text{分}$$

$$\lambda_2 = -3 \rightarrow \alpha_2 = [-1, 0, 1]^T, \quad \xi_2 = \frac{1}{\sqrt{2}} [-1, 0, 1]^T$$

$$\lambda_3 = -4 \rightarrow \alpha_3 = [1, -1, 1]^T, \quad \xi_3 = \frac{1}{\sqrt{3}} [1, -1, 1]^T$$

$$P = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix} \quad P^{-1} = \begin{bmatrix} \frac{1}{6} & \frac{1}{3} & \frac{1}{6} \\ -\frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{3} & -\frac{1}{3} & \frac{1}{3} \end{bmatrix} \rightarrow 4 \text{分}$$

$$(P^{-1})^T = (P^T)^{-1} = \begin{bmatrix} \frac{1}{6} & -\frac{1}{2} & \frac{1}{3} \\ \frac{1}{3} & 0 & -\frac{1}{3} \\ \frac{1}{6} & \frac{1}{2} & \frac{1}{3} \end{bmatrix} \rightarrow 6 \text{分}$$

$$\beta_1 = [\frac{1}{6} \quad \frac{1}{3} \quad \frac{1}{6}]^T \quad \beta_2 = [-\frac{1}{2} \quad 0 \quad \frac{1}{2}]^T \quad \beta_3 = [\frac{1}{3} \quad -\frac{1}{3} \quad \frac{1}{3}]^T$$

$$A = -1 \cdot G_1 + (-3) \cdot G_2 + (-4) \cdot G_3 \quad G_1 = \alpha_1 \beta_1^T = \begin{bmatrix} \frac{1}{6} & \frac{1}{3} & \frac{1}{6} \\ \frac{1}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{6} & \frac{1}{3} & \frac{1}{6} \end{bmatrix} \rightarrow 10 \text{分}$$

$$G_2 = \begin{bmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 0 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix} \quad G_3 = \alpha_3 \beta_3^T = \begin{bmatrix} \frac{1}{3} & -\frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} & -\frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

12分) 已知 Hermitian 矩阵 $A = \begin{bmatrix} 1 & 2-i & 0 \\ 2+i & 3 & 1 \\ 0 & 2+i & 2 \end{bmatrix}$, 与之相对应的 Hermitian 二次型为

$$f(X) = X^H A X, \text{ 这里 } X = [x_1, x_2, x_3]^T.$$

(1) 用酉变换将 Hermitian 二次型 $f(X) = X^H A X$ 化成标准形, 并写出所做的酉变换.

(2) 判断 $f(X) = X^H A X$ 的定性 (正定、负定, 半正定, 半负定, 不定).

$$f(\lambda) = |\lambda I - A| = (\lambda - 1)^2 (\lambda - 7) \quad \lambda_1 = \lambda_2 = 1, \lambda_3 = 7 \rightarrow 2 \text{分}$$

$$(\lambda_1 I - A)X = 0 \rightarrow \alpha_1 = [1, 0, 0]^T, \quad \alpha_2 = [0, 1, -2-i]^T$$

正交化单位化后可得 $\eta_1 = [1, 0, 0]^T, \quad \eta_2 = [0, \frac{1}{\sqrt{6}}, \frac{-2-i}{\sqrt{6}}]^T$
(= 正交) $\rightarrow 4 \text{分}$

$$\lambda_3 = 7, \quad (7I - A)X = 0, \quad \alpha_3 = [0, 2-i, 1]^T$$

单位化后 $\eta_3 = [0, \frac{2-i}{\sqrt{6}}, \frac{1}{\sqrt{6}}]^T \rightarrow 6 \text{分}$

$$U = [\eta_1, \eta_2, \eta_3] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{6}} & \frac{2-i}{\sqrt{6}} \\ 0 & -\frac{2-i}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix} \text{ 或 } \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{30}} & \frac{2-i}{\sqrt{30}} \\ 0 & \frac{2+i}{\sqrt{30}} & -\frac{1}{\sqrt{30}} \end{bmatrix}$$

$$U^H A U = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{bmatrix} \rightarrow 8 \text{分}$$

$$X = UY \text{ 则 } \lambda \quad f(X) = X^H A X, \text{ 记 } Y$$

$$f(X) = Y^H U^H A U Y = \bar{y}_1 y_1 + \bar{y}_2 y_2 + 7 \bar{y}_3 y_3. \rightarrow 10 \text{分}$$

由于 Hermitian 矩阵 A 的所有特征值都大于零，因此断定 $f(X) = X^H A X$ 为正定二次型 $\rightarrow 12 \text{分}$

六、(9分) 已知 $A = \begin{bmatrix} 4 & 0 & -2 \\ 0 & 6 & 0 \\ -1 & 0 & 5 \end{bmatrix}$ ，求 $\lim_{k \rightarrow \infty} \left(\frac{1}{\rho(A)} A \right)^k$ ，这里 $\rho(A)$ 表示矩阵 A 的谱半径。

$$f(\lambda) = |\lambda I - A| = (\lambda - 3)(\lambda - 6)^2, \quad \rho(A) = 6 \rightarrow 2 \text{分}$$

$$\lambda_1 = 3, \quad \lambda_2 = \lambda_3 = 6 \rightarrow 4 \text{分}$$

$$\downarrow \alpha_1 = [2, 0, 1]^T$$

$$\alpha_2 = [0, 1, 0]^T$$

$$\alpha_3 = [-1, 0, 1]^T$$

$$P = [\alpha_1, \alpha_2, \alpha_3] = \begin{bmatrix} 2 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \rightarrow 6 \text{分} \quad P^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 1 & 0 \\ -\frac{1}{3} & 0 & \frac{2}{3} \end{bmatrix}$$

$$P^{-1}AP = \begin{bmatrix} 3 & & \\ & 6 & \\ & & 6 \end{bmatrix} \quad \rho(A) = 6$$

$$\lim_{k \rightarrow \infty} (\frac{1}{\rho(A)} \cdot A)^k = \lim_{k \rightarrow \infty} \left(P \begin{bmatrix} \frac{1}{3} & & \\ & 1 & \\ & & 1 \end{bmatrix} P^{-1} \right)^k$$

$$= \begin{bmatrix} \frac{1}{3} & 0 & -\frac{2}{3} \\ 0 & 1 & 0 \\ -\frac{1}{3} & 0 & \frac{2}{3} \end{bmatrix} \rightsquigarrow \begin{bmatrix} \frac{1}{3} & 0 & -\frac{2}{3} \\ 0 & 1 & 0 \\ -\frac{1}{3} & 0 & \frac{2}{3} \end{bmatrix} \quad \rightsquigarrow \begin{bmatrix} \frac{1}{3} & 0 & -\frac{2}{3} \\ 0 & 1 & 0 \\ -\frac{1}{3} & 0 & \frac{2}{3} \end{bmatrix} \quad \rightsquigarrow \begin{bmatrix} \frac{1}{3} & 0 & -\frac{2}{3} \\ 0 & 1 & 0 \\ -\frac{1}{3} & 0 & \frac{2}{3} \end{bmatrix}$$

七、(10分) 已知矩阵 $A = \begin{bmatrix} 2 & 6 \\ 2 & 3 \end{bmatrix}$. 证明: 矩阵幂级数 $\sum_{k=1}^{\infty} \frac{k}{10^k} A^k$ 收敛, 并求其收敛和.

$$\sum_{k=1}^{\infty} \frac{k}{10^k} A^k = \sum_{k=1}^{\infty} k \cdot \left(\frac{A}{10} \right)^k \quad B = \frac{A}{10} = \begin{bmatrix} \frac{1}{5} & \frac{3}{5} \\ \frac{1}{5} & \frac{3}{10} \end{bmatrix}$$

$$|\lambda I - B| = (\lambda - \frac{3}{5})(\lambda + \frac{1}{10}). \quad \rho(B) = \frac{3}{5} \rightsquigarrow 4 \text{ 分}$$

$$\frac{1}{\rho} = \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = 1 > \rho(B). \quad \text{故 } \sum_{k=1}^{\infty} \frac{k}{10^k} A^k \text{ 收敛.}$$

$$\sum_{k=1}^{\infty} k x^k = x(1-x)^{-2} \rightsquigarrow 8 \text{ 分}$$

$$\sum_{k=1}^{\infty} \frac{k}{10^k} A^k = \frac{A}{10} (I - \frac{A}{10})^{-2} \rightsquigarrow 9 \text{ 分}$$

$$= \begin{bmatrix} \frac{755}{484} & \frac{795}{242} \\ \frac{265}{242} & \frac{255}{121} \end{bmatrix} \rightsquigarrow 10 \text{ 分}$$

八、(5分) 设 $M_{3,3}(R)$ 为实数域 R 上所有三阶矩阵构成的线性空间, $W = \{A \in M_{3,3}(R) | \text{Tr}(A) = 0\}$,

这里 $\text{Tr}(A)$ 表示矩阵 A 的迹, 求 W 的维数, 并写出 W 的一组基.

$$\dim W = 8 \rightsquigarrow 2\frac{1}{2}$$

一组基为

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$\rightsquigarrow 3\frac{1}{2}$$